

Section : Chemistry

Q.1 Consider the following statements for the order of a reaction and choose the correct statements.

- (A). The order of a reaction is an experimental quantity.
 (B). Order can be zero.
 (C). Order can be fractional value.
 (D). Order is applicable to elementary as well as complex reactions.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
 2. (A), (B) and (C) only
 3. (A), (B), (C) and (D)
 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Match the complex given in List-I with its property given in List-II

List-I	List-II
Complex	Property

(A) $[\text{CoF}_6]^{3-}$ (I) d^5 , high spin

(B) $[\text{Fe}(\text{CN})_6]^{4-}$ (II) d^6 , high spin

(C) $[\text{FeF}_6]^{3-}$ (III) d^5 , low spin

(D) $[\text{Mn}(\text{CO})_6]^{2+}$ (IV) d^6 , low spin

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
 2. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
 3. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
 4. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.3 Arrange the following in increasing order of their basic strength in aqueous phase :

(A) CH_3NH_2

(B) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

(C) $\text{C}_6\text{H}_5\text{NH}_2$

(D) $(\text{CH}_3)_3\text{N}$

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
 2. (A), (C), (B), (D)
 3. (B), (A), (D), (C)
 4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Match List-I with List-II

List-I	List-II
Cell	Reaction

(A) Fuel Cell (I) Anode: $\text{Zn}(s) \rightarrow \text{Zn}^{2+} + 2e^-$
 Cathode: $\text{MnO}_2 + \text{NH}_4^+ + e^- \rightarrow \text{MnO}(\text{OH}) + \text{NH}_3$

(B) A secondary cell (II) Cathode: $\text{O}_2(g) + 2\text{H}_2\text{O}(l) + 4e^- \rightarrow 4\text{OH}^-(aq)$
 Anode: $2\text{H}_2(g) + 4\text{OH}^-(aq) \rightarrow 4\text{H}_2\text{O}(l) + 4e^-$

(C) Mercury cells (III) $\text{Pb}(s) + \text{PbO}_2(s) + 2\text{H}_2\text{SO}_4(aq) \rightarrow 2\text{PbSO}_4(s) + 2\text{H}_2\text{O}(l)$

(D) Leclanche cell (IV) $\text{Zn(Hg)} + \text{HgO}(s) \rightarrow \text{ZnO}(s) + \text{Hg}(l)$
(Simplified)

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Match List-I with List-II with the corresponding boiling points.

List-I	List-II
Compound Name	Boiling Point
(A) $\text{CH}_3\text{CH}_2\text{OH}$	(I) 373K
(B) CH_3OCH_3	(II) 348K
(C) H_2O	(III) 351K
(D) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$	(IV) 248K

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
2. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.6 What product(s) is formed when an Alkyl halide undergoes ammonolysis ?

- (A) Primary amine
- (B) Secondary amine
- (C) Quaternary ammonium salt
- (D) Tertiary amine

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) (C) and (D)
3. (B), (C) and (D) only
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.7 What is the formula of following coordination compound:
Potassium trioxalatoaluminate(III)

1. $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$
2. $\text{K}[\text{Al}_2(\text{C}_2\text{O}_4)_3]$
3. $\text{K}_3[\text{Al}_2(\text{C}_2\text{O}_4)_3]$
4. $\text{K}[\text{Al}(\text{C}_2\text{O}_4)_3]$

Options 1. 1

2. 2
3. 3
4. 4

Q.8 What is the structural formula for Freon-12.

1. CCl_3F
2. CCl_2F_2
3. CClF_3
4. CCl_4

Options 1. 1

2. 2
3. 3
4. 4

Q.9 "Soft drink bottles are sealed under high pressure to increase the solubility of carbon dioxide" is the application of which of the following laws:

1. Dalton's law
2. Henry's law
3. Charles's law
4. Boyle's law

Options

1. 1
2. 2
3. 3
4. 4

Q.10 Which one of the following statements is correct for a first order reaction?

1. Half life is directly proportional to the initial concentration of reactant.
2. Half life is inversely proportional to the initial concentration of reactant.
3. Infinite time is taken for 100% completion of the reaction.
4. Rate of the reaction is proportional to the square of initial concentration of reactant.

Options

1. 1
2. 2
3. 3
4. 4

Q.11 Match List-I with List-II

List-I List-II

Type of solution Example

(A) solid in solid (I) Camphor in N₂

(B) solid in gas (II) Cu in Au

(C) gas in solid (III) Hg in Na

(D) liquid in solid (IV) H₂ in Pd

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.12 What is Lucas Reagent?

1. FeCl₃ and conc. HCl
2. CuCl₂ and conc. HCl
3. ZnCl₂ and conc. HCl
4. MgCl₂ and conc. HCl

Options

1. 1
2. 2
3. 3
4. 4

Q.13 Williamson's synthesis for preparing dimethyl ether involves

1. S_N1 mechanism
2. Elimination reaction
3. S_N2 mechanism
4. Nucleophilic addition reaction

Options

1. 1
2. 2
3. 3
4. 4

Q.14 What is the IUPAC name of m-cresol?

1. 3-Methylphenol
2. 3-Chlorophenol
3. 3-Methoxyphenol
4. Benzene-1,3-diol

Options

1. 1
2. 2

3. 3

4. 4

Q.15 Match List-I with List-II

List-I List-II

Amino acid R group

(A). Alanine (I). -CH₃

(B). Lysine (II). HO-CH₂-

(C). Serine (III). H₂N-(CH₂)₄-

(D). Cysteine (IV). HS-CH₂-

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.16 Identify the least basic amine in gaseous phase?

1. (CH₃)₂NH

2. CH₃CH₂NH₂

3. C₆H₅-NH₂

4. (CH₃)₃N

Options 1. 1

2. 2

3. 3

4. 4

Q.17 If an electrolyte on dissociation gives ν_+ cations and ν_- anions, then its limiting molar conductivity is given by

1. $\Lambda_m^0 = \nu_- \lambda_+^0 - \nu_+ \lambda_-^0$

2. $\Lambda_m^0 = \nu_+ \lambda_+^0 - \nu_- \lambda_-^0$

3. $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$

4. $\Lambda_m^0 = \nu_+ \lambda_+^0 + \nu_- \lambda_-^0$

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Select the correct relation of half life for a zero order reaction.

1. $t_{0.5} = \frac{[R]_0}{2k}$

2. $t_{0.5} = \frac{2k}{[R]_0}$

3. $t_{0.5} = \frac{[R]_0}{k}$

4. $t_{0.5} = \frac{[R]_0}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Aryl halides are extremely less reactive towards nucleophilic substitution reaction because of

1. Electromeric effect

2. Resonance effect

3. Inductive effect

4. Hyperconjugation effect.

Options 1. 1

2. 2

3. 3

- Q.20** The conductivity of electrolytic (ionic) solution depends on
- (A). Nature of electrolyte added
 - (B). Nature of solvent
 - (C). Temperature
 - (D). Size of ions produced

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.21** The thermal decomposition of HI on gold surface is an example of

1. First order reaction
2. Zero order reaction
3. Second order reaction
4. Pseudo first order reaction

Options

1. 1
2. 2
3. 3
4. 4

- Q.22** Which of the following is NOT a disaccharide?

1. Sucrose
2. Maltose
3. Lactose
4. Cellulose

Options

1. 1
2. 2
3. 3
4. 4

- Q.23** All natural and artificial radioactive decay of unstable nuclei take place by

1. Zero order kinetics
2. First order kinetics
3. Second order kinetics
4. Third order kinetics

Options

1. 1
2. 2
3. 3
4. 4

- Q.24** Reduction of nitrobenzene by which of the following reagent does not give aniline?

1. Sn/HCl
2. Fe/HCl
3. H₂-Pd
4. Sn/NH₄OH

Options

1. 1
2. 2
3. 3
4. 4

- Q.25** The solute and solvent in the solution of hydrogen in palladium are

1. Solute: Gas; Solvent: Solid
2. Solute: Gas; Solvent: Liquid
3. Solute: Liquid; Solvent: Gas
4. Solute: Liquid; Solvent: Solid

Options

1. 1
2. 2
3. 3
4. 4

- Q.26** The synthesis of Alkyl fluoride is best carried through :

1. Free radicals
2. Swarts Reaction
3. Sandmeyer Reaction
4. Finkelstein Reaction

Options

1. 1

2. 2
3. 3
4. 4

Q.27 Based on CFT, in octahedral crystal field, d orbitals split into orbitals of t_{2g} and e_g sets. Which d orbitals forms the t_{2g} set?

1. d_{xy}, d_{yz}, d_{zx}
2. d_{xy}, d_{yz}, d_{z^2}
3. $d_{x^2}, d_{x^2-y^2}$
4. $d_{xy}, d_{x^2-y^2}$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.28 What is the value of van't Hoff factor for complete dissociation of K_2SO_4 ?

1. 1.00
2. 2.00
3. 3.00
4. 4.00

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.29 Which of the following compound will undergo azo coupling reaction with benzene diazonium chloride?

- (A) Aniline
- (B) Phenol
- (C) p-Toluidine
- (D) Nitrobenzene

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (A), (B) and (C) only

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.30 Which of the followings is NOT a monosaccharide?

1. Glucose
2. Fructose
3. Ribose
4. Sucrose

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.31 Arrange the following electrodes in increasing order of strength as an oxidizing agent.

- (A) $Ni^{2+} + 2e^- \rightarrow Ni(s) \quad E^0 = -0.25 \text{ V}$
- (B) $Cu^{2+} + 2e^- \rightarrow Cu(s) \quad E^0 = 0.34 \text{ V}$
- (C) $Br_2 + 2e^- \rightarrow 2Br^- \quad E^0 = 1.09 \text{ V}$
- (D) $O_2(g) + 2H^+ + 2e^- \rightarrow H_2O_2 \quad E^0 = 0.68 \text{ V}$

Choose the correct answer from the options given below:

1. (C), (D), (B), (A)
2. (B), (A), (C), (D)
3. (B), (A), (D), (C)
4. (A), (B), (D), (C)

- Options
1. 1
 2. 2

3. 3

4. 4

Q.32 Choose the correct sequence for the increasing boiling point of the following compounds.

- (A) Pentan-1-ol
- (B) Butan-2-ol
- (C) Ethanol
- (D) Propan-1-ol

Choose the correct answer from the options given below:

- 1. (C), (D), (B), (A)
- 2. (C), (D), (A), (B)
- 3. (A), (D), (B), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.33 Arrange the following ligands in increasing order of their field strength:

- (A) SCN-
- (B) F-
- (C) H₂O
- (D) CO

Choose the correct answer from the options given below:

- 1. (B), (C), (D), (A)
- 2. (C), (A), (B), (D)
- 3. (A), (B), (C), (D)
- 4. (A), (C), (D), (B)

Options 1. 1

2. 2

3. 3

4. 4

Q.34 What is the IUPAC name of Glycerol?

- 1. Propane-1,2,3-triol
- 2. Butane-1,2,3-triol
- 3. Ethane-1,2-diol
- 4. Propane-1,2,-diol

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Which of following reagents will furnish ambident nucleophile?

- (A) AgCN
- (B) KCN
- (C) AgNO₂
- (D) KNO₂

Choose the correct answer from the options given below:

- 1. (A), (B), (C) and (D)
- 2. (A), (B) and (C) only
- 3. (A), (B), and (D) only
- 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.36 Select the correct combination of coordination number and type of hybridization, respectively, in "trigonal bipyramidal" complex

- 1. 4 and sp^3d
- 2. 5 and sp^3d
- 3. 5 and dsp^3
- 4. 4 and dsp^2

Options 1. 1

2. 2

3. 3

4. 4

Q.37 The unit of molar conductivity is

1. S cm mol^{-1}
2. $\text{S cm}^2 \text{ mol}^{-2}$
3. $\text{ohm cm}^2 \text{ mol}^{-1}$
4. $\text{S cm}^2 \text{ mol}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.38 A solution of 2.5 g of ethanoic acid in 75 g of benzene is prepared. What is the molar mass of ethanoic acid, moles of ethanoic acid, mass of benzene and molality of ethanoic acid?

- (A) $0.556 \text{ mol kg}^{-1}$
- (B) 0.0417 mol
- (C) 60 g mol^{-1}
- (D) 0.075 kg

Choose the correct answer from the options given below:

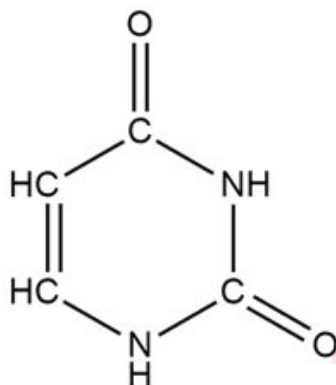
1. (A), (B), (C), (D)
2. (A), (B), (D), (C)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.39

The following structure of nucleic acid base is



1. Uracil
2. Thymine
3. Cytosine
4. Guanine

Options 1. 1

2. 2
3. 3
4. 4

Q.40 Which of the following base is NOT present in DNA?

1. Adenine
2. Guanine
3. Cytosine
4. Uracil

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer the following questions

There are mainly four series of the transition metals, *3d* series (Sc to Zn), *4d* series (Y to Cd), *5d* series (La and Hf to Hg) and *6d* series which has Ac and elements from Rf to Cn. The two series of the inner transition

metals; $4f$ (Ce to Lu) and $5f$ (Th to Lr) are known as lanthanoids and actinoids respectively. Their chemical properties are transitional between those of s and p -block elements. The transition metals (with the exception of Zn, Cd and Hg) are very hard and have low volatility. Their melting and boiling points are high. Transition metals have incomplete d subshell either in neutral atom or in their ions. With partly filled d orbitals these elements exhibit certain characteristic properties such as display of a variety of oxidation states, formation of coloured ions and entering into complex formation with a variety of ligands. The transition metals and their compounds are known for their catalytic activity.

Which of the following statements is correct?

1. All transition metals have high volatility.
2. Due to an increase in nuclear charge, there is an increase in the ionization enthalpy along the transition series from left to right.
3. The high melting points of transition metals are attributed to the involvement of fewer number of electrons in the interatomic metallic bonding.
4. Shielding effect of d -electrons is less than that of $4f$ electrons

Options

- 1.
- 2.
- 3.
- 4.

Q.42 Read the passage carefully and answer the following questions

There are mainly four series of the transition metals, $3d$ series (Sc to Zn), $4d$ series (Y to Cd), $5d$ series (La and Hf to Hg) and $6d$ series which has Ac and elements from Rf to Cn. The two series of the inner transition metals; $4f$ (Ce to Lu) and $5f$ (Th to Lr) are known as lanthanoids and actinoids respectively. Their chemical properties are transitional between those of s and p -block elements. The transition metals (with the exception of Zn, Cd and Hg) are very hard and have low volatility. Their melting and boiling points are high. Transition metals have incomplete d subshell either in neutral atom or in their ions. With partly filled d orbitals these elements exhibit certain characteristic properties such as display of a variety of oxidation states, formation of coloured ions and entering into complex formation with a variety of ligands. The transition metals and their compounds are known for their catalytic activity.

In series Sc to Zn, the lowest enthalpy of atomization is shown by

1. Sc
2. Zn
3. Mn
4. Co

Options

- 1.
- 2.
- 3.
- 4.

Q.43 Read the passage carefully and answer the following questions

There are mainly four series of the transition metals, $3d$ series (Sc to Zn), $4d$ series (Y to Cd), $5d$ series (La and Hf to Hg) and $6d$ series which has Ac and elements from Rf to Cn. The two series of the inner transition metals; $4f$ (Ce to Lu) and $5f$ (Th to Lr) are known as lanthanoids and actinoids respectively. Their chemical properties are transitional between those of s and p -block elements. The transition metals (with the exception of Zn, Cd and Hg) are very hard and have low volatility. Their melting and boiling points are high. Transition metals have incomplete d subshell either in neutral atom or in their ions. With partly filled d orbitals these elements exhibit certain characteristic properties such as display of a variety of oxidation states, formation of coloured ions and entering into complex formation with a variety of ligands. The transition metals and their compounds are known for their catalytic activity.

In general the electronic configuration of outer orbitals of transition metals is $(n-1)d^{1-10}ns^{1-2}$ but Pd shows an

exceptional electronic configuration as.....

1. $4d^5 5s^0$
2. $4d^{10} 5s^0$
3. $4d^5 5s^1$
4. $4d^{10} 5s^1$

Options

- 1.

2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer the following questions

There are mainly four series of the transition metals, *3d* series (Sc to Zn), *4d* series (Y to Cd), *5d* series (La and Hf to Hg) and *6d* series which has Ac and elements from Rf to Cn. The two series of the inner transition metals; *4f* (Ce to Lu) and *5f* (Th to Lr) are known as lanthanoids and actinoids respectively. Their chemical properties are transitional between those of *s* and *p*-block elements. The transition metals (with the exception of Zn, Cd and Hg) are very hard and have low volatility. Their melting and boiling points are high. Transition metals have incomplete *d* subshell either in neutral atom or in their ions. With partly filled *d* orbitals these elements exhibit certain characteristic properties such as display of a variety of oxidation states, formation of coloured ions and entering into complex formation with a variety of ligands. The transition metals and their compounds are known for their catalytic activity.

The highest oxidation states shown by V, Fe and Ni respectively are

1. +6, +3, +2
2. +5, +6, +4
3. +4, +5, +2
4. +5, +3, +2

Options

1. 1
2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer the following questions

There are mainly four series of the transition metals, *3d* series (Sc to Zn), *4d* series (Y to Cd), *5d* series (La and Hf to Hg) and *6d* series which has Ac and elements from Rf to Cn. The two series of the inner transition metals; *4f* (Ce to Lu) and *5f* (Th to Lr) are known as lanthanoids and actinoids respectively. Their chemical properties are transitional between those of *s* and *p*-block elements. The transition metals (with the exception of Zn, Cd and Hg) are very hard and have low volatility. Their melting and boiling points are high. Transition metals have incomplete *d* subshell either in neutral atom or in their ions. With partly filled *d* orbitals these elements exhibit certain characteristic properties such as display of a variety of oxidation states, formation of coloured ions and entering into complex formation with a variety of ligands. The transition metals and their compounds are known for their catalytic activity.

Which of the following is not regarded as transition metal?

1. Mercury (Hg)
2. Silver (Ag)
3. Technitium (Tc)
4. Scandium (Sc)

Options

1. 1
2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the following questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

Which of the following compounds will give butanone on oxidation with chromic anhydride (CrO₃)?

1. Butan-1-ol
2. Butan-2-ol
3. 2-Methyl propane
4. 2-Methyl butane

Options

1. 1
2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the following questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

Identify the incorrect statement for Cannizaro reaction.

1. The reaction involves self oxidation and reduction
2. It is given by aldehydes which lack alpha hydrogens
3. It is carried out in presence of concentrated alkali
4. Dilute sodium hydroxide is used as a catalyst

Options

1. 1
2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the following questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

The IUPAC name of following compound is

$\text{OHC-C}_6\text{H}_4\text{-CHO-p}$

1. Benzene-1,4-dial
2. 1,4-Dioxobenzene
3. 4-Formylbenzaldehyde
4. Benzene-1,4-dicarbaldehyde

Options

1. 1
2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the following questions

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Arrange the following in the increasing order of their pK_a values:

- (A) $\text{CF}_3\text{-COOH}$
- (B) $\text{O}_2\text{N-CH}_2\text{-COOH}$
- (C) $\text{NC-CH}_2\text{-COOH}$
- (D) H-COOH

Choose the correct answer from the options given below:

- 1. (D), (C), (B), (A)
- 2. (D), (C), (A), (B)
- 3. (A), (C), (B), (D)
- 4. (D), (B), (C), (A)

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.50 Read the passage carefully and answer the following questions

Aldehydes, carboxylic acids and ketones are widespread in the animal and plant kingdoms. They play a pivotal role in various biochemical processes of life. Their presence in nature adds fragrance and flavor. These compounds are widely used in food products, pharmaceuticals, paints, resins and other important product industries. These compounds are prepared by various laboratory methods, which mainly include oxidation, formylation, acylation and reduction. Due to the polar nature of the carbonyl group in aldehydes and ketones, they can exhibit different reactions like nucleophilic addition. They do exhibit redox and various condensation reactions which lead to the formation of various important compounds. On the other hand, the carboxylic acids are mainly prepared by oxidation and hydrolysis of different compounds. The carboxylic acid consists of a carbonyl group and the hydroxy group (attached to the carbonyl carbon atom). This makes it possible for the carboxylic acid to participate in various chemical reactions which involve cleavage of the C-OH bond and the O-H bond along with the reactions involving the complete -COOH group.

The reagent which doesn't react with both acetone and benzaldehyde is:

- 1. Sodium hydrogensulphite
- 2. Phenyl hydrazine
- 3. Fehling's solution
- 4. Grignard reagent

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4